Joint Longitudinal-Survival Analysis of Postoperative Left Ventricular Mass Index (LVMI) and Mortality After Aortic Valve Replacement

Ryan Kitagawa, Linlin Liu, and Nancy Lopez
University of California, Riverside
STAT 293
March 12, 2026

Table of Contents

1. Introduction
2. Exploratory Data Analysis
3. Modeling
4. Conclusion
5. Contribution
6. References

Table of Contents

1. Introduction
2. Exploratory Data Analysis
3. Modeling
4. Conclusion
5. Contribution
6. References

Clinical Context

- Aortic valve replacement is a common surgical treatment for severe aortic valve disease
- After surgery, patients are followed longitudinally using echocardiographic measurements to monitor cardiac recovery
- Left ventricular mass index (LVMI) is an important biomarker reflecting cardiac remodeling and regression of ventricular hypertrophy

Statistical Problem

- Repeated longitudinal measurements of LVMI, along with other outcomes, are recorded at multiple follow-up visits after surgery
- At the same time, patients are followed for time-to-event outcomes, such as death or major complications
- These two processes are observed on the same patients over time and may be related

Limitations of Classical Models

- Linear mixed-effects models (LMMs) analyze longitudinal biomarker trajectories but ignore survival outcomes
- Survival models like Cox proportional hazards (PH) model analyze time-to-event outcomes but treat biomarkers as fixed covariates and ignore their longitudinal evolution

Limitations of Classical Models

- Linear mixed-effects models (LMMs) analyze longitudinal biomarker trajectories but ignore survival outcomes
- Survival models like Cox proportional hazards (PH) model analyze time-to-event outcomes but treat biomarkers as fixed covariates and ignore their longitudinal evolution
- As a result, analyzing these processes separately may lead to inefficient or biased inference when the biomarker process and survival process are related

Data Structure

- Observational study of patients undergoing aortic valve replacement surgery
- 988 total follow-ups from 256 unique patients
- Repeated longitudinal measurements of cardiac function recorded from the date of surgery
  - **LVMI (missingness) and log(LVMI) (no missingness)**
  - Valve gradient
  - Ejection fraction
- Survival status recorded at the maximum follow-up visit from the date of surgery

Key Variables

- `num` – patient identifier
- `log.lvmi` – natural log transformation of LVMI at follow-up visit
- `time` – observed time point, with surgery date as the time origin (years)
- `fuyrs` – maximum follow-up time, with surgery date as the time origin (years)
- `status` – censoring indicator (1 = died, 0 = lost at follow-up)

Baseline Covariates

- `sex` – patient sex (0 = Male, 1 = Female)
- `age` – age at surgery (years)
- `bsa` – preoperative body surface area
- `lvh` – preoperative left ventricular hypertrophy
- `prenyha` – preoperative New York Heart Association (NYHA) classification (1 = I/II, 3 = III/IV)
- `redo` – previous cardiac surgery
- `dm` – preoperative diabetes
- `lv` – left ventricular ejection fraction (1 = good, 2 = moderate, 3 = poor)
- `hs` – valve type (1 = homograft, 0 = stentless porcine)

Our Approach

Primary question

How are the postoperative trajectories of left ventricular mass index (LVMI) following aortic valve replacement associated with time to death?

Primary model

- We apply a joint longitudinal-survival model to simultaneously analyze LVMI trajectories and mortality
  - Longitudinal submodel: postoperative LVMI trajectory
  - Survival submodel: time to death

Our Approach

Primary question

How are the postoperative trajectories of left ventricular mass index (LVMI) following aortic valve replacement associated with time to death?

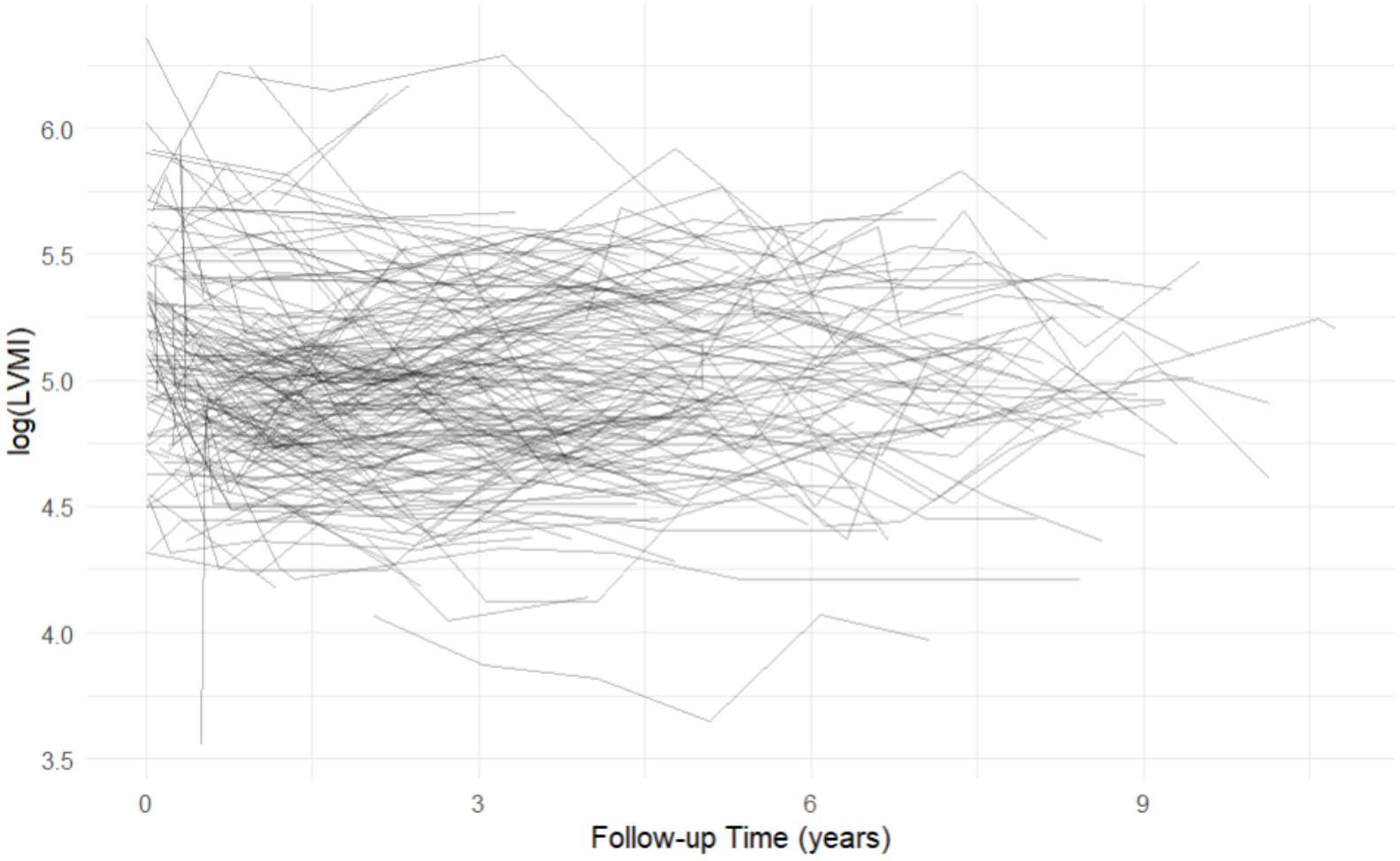
Primary model

- We apply a joint longitudinal-survival model to simultaneously analyze LVMI trajectories and mortality
  - Longitudinal submodel: postoperative LVMI trajectory
  - Survival submodel: time to death
  - Accounts for within-subject correlation and informative dropout due to mortality

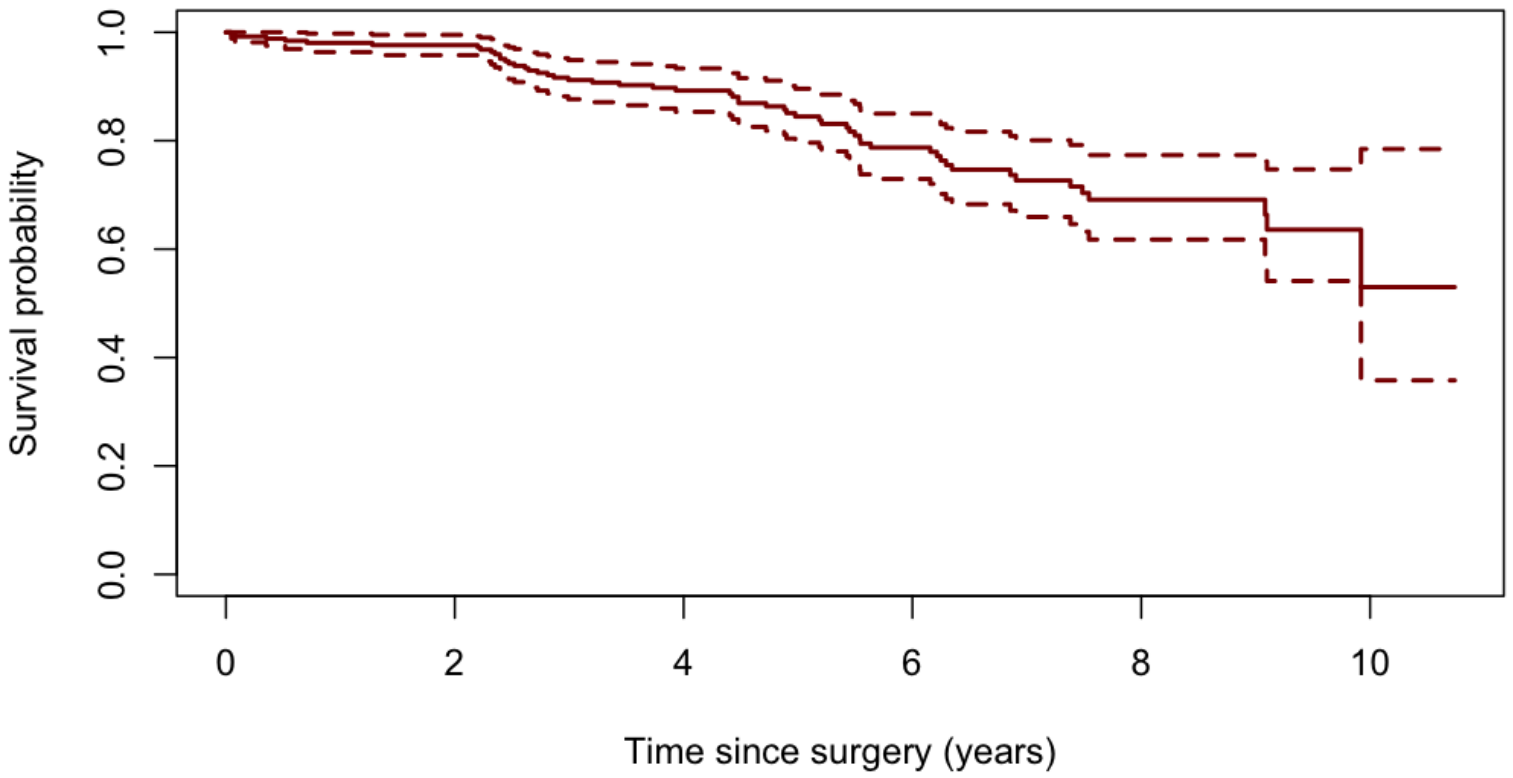
Table of Contents

1. Introduction
2. Exploratory Data Analysis
3. Modeling
4. Conclusion
5. Contribution
6. References

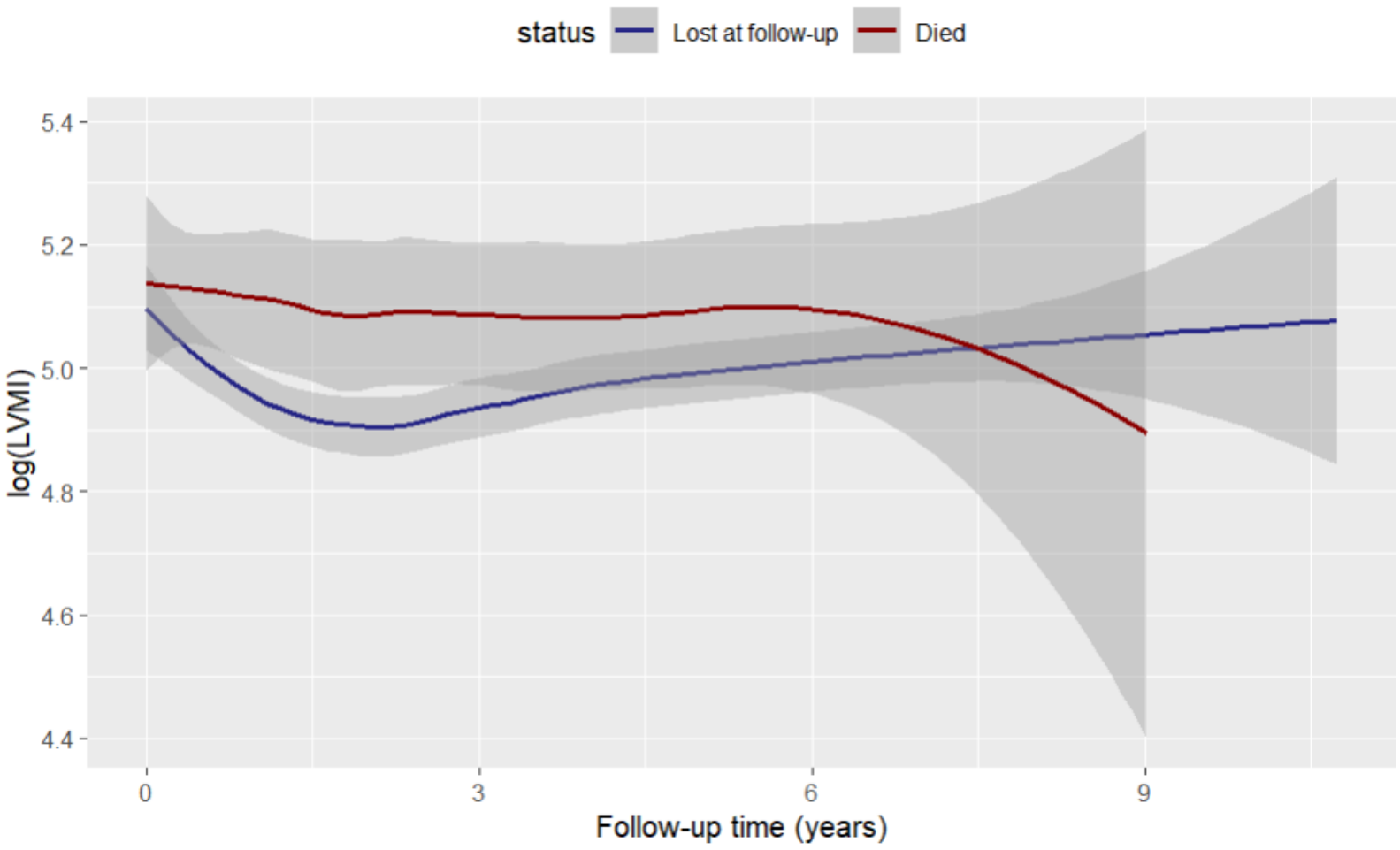
Spaghetti Plot of log(LVMI) Trajectories Over Time



Overall Kaplan-Meier Survival Curve



Smoothed log(LVMI) Trajectories by Survival Status



Slope/Rate of Change Parameterization

- EDA of smoothed `log.lvmi` trajectories suggested differences in postoperative trends between the two survival status
- Clinically, two patients with the same LVMI value may have different prognoses if one is improving while the other is worsening

Slope/Rate of Change Parameterization

- EDA of smoothed `log.lvmi` trajectories suggested differences in postoperative trends between the two survival status
- Clinically, two patients with the same LVMI value may have different prognoses if one is improving while the other is worsening
- Extend the joint model so the hazard depends not only on the LVMI level $m_i(t)$, but also on its instantaneous rate of change $m_i'(t)$, to assess whether dynamic LVMI progression carries additional prognostic information beyond the current LVMI level

Table of Contents

1. Introduction
2. Exploratory Data Analysis
3. Modeling
4. Conclusion
5. Contribution
6. References

Longitudinal Submodel

Linear mixed-effects model (LMM)

$$Y_{ij} = m_i(t_{ij}) + \varepsilon_{ij}, \quad \varepsilon_{ij} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2),$$

$$m_i(t) = \beta_0 + \beta_1 t + \boldsymbol{\beta}^\top \mathbf{Z}_i + b_{0i} + b_{1i} t,$$

where

- Y_{ij} represents the observed LVMI measurement for subject i at time t_{ij},
- $\mathbf{Z}_i$ contains baseline covariates, and
- b_{0i}, b_{1i} represent subject-specific random intercepts and slopes.

LMM: Assumptions and Diagnostics

- Residuals vs. fitted plots to assess homoscedasticity and model fit
- Q-Q plots of residuals to evaluate normality of error terms
- Q-Q plots of random intercepts and slopes to assess normality of random effects
- Scatter plot of random intercepts vs. slopes to examine random effects structure
- Likelihood comparison with quadratic time term to assess potential nonlinearity in time

Survival Submodel

Cox Proportional Hazards (PH) Model

$$h_i(t) = h_0(t)\exp(\boldsymbol{\gamma}^\top \mathbf{X}_i),$$

where

- $h_i(t)$ is the hazard of death at time t for subject i,
- $h_0(t)$ is the baseline hazard,
- $\mathbf{X}_i$ represents baseline covariates, and
- $m_i(t)$ is the latent true LVMI trajectory from the longitudinal model.

Cox PH: Assumptions and Diagnostics

- Schoenfeld residual tests and plots to assess the PH assumption
- Martingale residual plots to evaluate the functional form of continuous predictor age

Joint Longitudinal-Survival Model

Survival Model with Longitudinal Association

$$h_i(t) = h_0(t) \exp\left(\boldsymbol{\gamma}^\top \mathbf{X}_i + \alpha m_i(t)\right),$$

where

- $m_i(t)$ is the latent LVMI trajectory from the LMM, and
- α quantifies the association between LVMI and mortality risk.

The two submodels are connected through shared random effects.

Base Joint Model

- LMM: `time`, `hs`, `sex`, `age`, `bsa`, `lvh`, `prenyha`, `redo`, `dm`, and `lv`
- Cox PH: `hs`, `sex`, `age`, `bsa`, `lvh`, `prenyha`, `redo`, `dm`, `lv`, and `cluster(num)`
- `jointModel()` in `JM` R package
  - `timeVar = "time"`
  - `method = "piecewise-PH-aGH"`

Base Joint Model: Results

Log Likelihood	AIC	BIC
-282.4534	632.9069	753.4429

Table: Model Diagnostics

	Standard Deviation	Correlation
Intercept	0.3072	(Intr)
Time	0.0417	-0.3557
Residual	0.1928	

Table: Variance Components

$$\alpha \approx 1.25, \text{SE} = 0.5825, p = 0.0313$$

LMM Covariates Selection and Cox PH Stepwise

- LMM: `time`, `hs`, `sex`, `age`, `bsa`, `lvh`, `dm`, and `lv`
- Cox PH stepwise: Backward AIC $\rightarrow$ `age`, `lv`, and `cluster(num)`

Joint Model 2: Results

Log Likelihood	AIC	BIC
-285.5238	621.0476	709.6770

Table: Model Diagnostics

	Standard Deviation	Correlation
Intercept	0.3072	(Intr)
Time	0.0418	-0.3558
Residual	0.1928	

Table: Variance Components

$$\alpha \approx 0.99, \text{SE} = 0.5458, p = 0.0696$$

Joint Model 3

- LMM: `time`, `hs`, `sex`, `age`, `bsa`, `lvh`, `dm`, and `lv`
- Cox PH: `age`, I(`lv == 3`), and `cluster(num)`

Joint Model 3: Results

Log Likelihood	AIC	BIC
-286.2885	620.5770	705.6612

Table: Model Diagnostics

	Standard Deviation	Correlation
Intercept	0.3072	(Intr)
Time	0.0418	-0.3557
Residual	0.1928	

Table: Variance Components

$$\alpha \approx 1.06, \text{SE} = 0.5425, p = 0.0507$$

Joint Model with Slope Parameterization

- The current-value joint model links mortality risk to the level of the latent LVMI trajectory $m_i(t)$
- Two patients can have the same LVMI value but different prognoses if one is improving while the other is worsening
- Lim et al. (2008) pointed out the difference in early stage LVMI trajectory
- Extend the joint model by allowing the hazard to depend on both the current LVMI level and its instantaneous rate of change:

$$h_i(t) = h_0(t) \exp\left(\boldsymbol{\gamma}^\top \mathbf{X}_i + \alpha_1 m_i(t) + \alpha_2 m_i'(t)\right)$$

Implementation

Implementation

- Removed duplicate time points within subjects and excluded measurements occurring after the event or censoring time

Implementation

- Removed duplicate time points within subjects and excluded measurements occurring after the event or censoring time
- Restricted the analysis to individuals with complete covariate information

Implementation

- Removed duplicate time points within subjects and excluded measurements occurring after the event or censoring time
- Restricted the analysis to individuals with complete covariate information
- Refit the latest LMM and Cox PH survival submodels using the cleaned dataset

Implementation

- Removed duplicate time points within subjects and excluded measurements occurring after the event or censoring time
- Restricted the analysis to individuals with complete covariate information
- Refit the latest LMM and Cox PH survival submodels using the cleaned dataset
- `jointModel()`
  - `timeVar = "time"`
  - `method = "piecewise-PH-aGH"`
  - `parameterization = "slope"`
  - `derivForm = list()` for derivative specification

Results

Log Likelihood	AIC	BIC
-287.8694	621.7388	703.2778

Table: Model Diagnostics

	Standard Deviation	Correlation
Intercept	0.3138	(Intr)
Time	0.0433	-0.4075
Residual	0.1923	

Table: Variance Components

$$\alpha \approx -6.15, \text{SE} = 6.2488, p = 0.3247$$

Sensitivity Analysis: Extending Cox PH Submodel

- Expanded the survival submodel to additionally adjust for `sex` and `hs`
- Did not show significant association with mortality
- Overall model fit did not improve (AIC = 624.3 vs. 620.6 in the joint model 3)
- The association between current LVMI and mortality remained positive and statistically significant ($\alpha = 1.16$, p = 0.041)

Sensitivity Analysis: Value and Slope Parameterization

- Evaluated a joint model including both the current value and slope of LVMI
- Model fit was nearly identical to the value-only model (AIC $= 620.3$ vs. 620.6)
- The current value of LVMI remained positively and significantly associated with mortality ($\alpha = 1.25$, p $= 0.021$), while the slope effect was not statistically significant (p $= 0.28$)

Table of Contents

1. Introduction
2. Exploratory Data Analysis
3. Modeling
4. Conclusion
5. Contribution
6. References

Conclusion

Conclusion

- The joint longitudinal-survival model indicated a positive association between LVMI level and mortality risk

Conclusion

- The joint longitudinal-survival model indicated a positive association between LVMI level and mortality risk
- Incorporating a slope parameterization did not improve model fit and was not statistically significant

Conclusion

- The joint longitudinal-survival model indicated a positive association between LVMI level and mortality risk
- Incorporating a slope parameterization did not improve model fit and was not statistically significant
- Results were consistent across multiple model specifications and sensitivity analyses

Conclusion

- The joint longitudinal-survival model indicated a positive association between LVMI level and mortality risk
- Incorporating a slope parameterization did not improve model fit and was not statistically significant
- Results were consistent across multiple model specifications and sensitivity analyses
- Future work could explore more flexible longitudinal structures, such as nonlinear LVMI trajectories

Thank You!

Questions?

Table of Contents

1. Introduction
2. Exploratory Data Analysis
3. Modeling
4. Conclusion
5. Contribution
6. References

	EDA	Primary Model Building	Extension Model Building	Presentation Slides
R. K.	✓	✓		✓
L. L.			✓	✓
N. L.	✓	✓		✓

Table of Contents

1. Introduction
2. Exploratory Data Analysis
3. Modeling
4. Conclusion
5. Contribution
6. References

References

- Hickey, G. L., et al. (2016). Joint modelling of time-to-event and multivariate longitudinal outcomes: recent developments and issues. *BMC Medical Research Methodology*. 16(1):117.
- Li, W., Li, L., & Astor, B. C. (2023). A comparison of two approaches to dynamic prediction: joint modeling and landmark modeling. *Statistics in Medicine*. 42(13):2101-2115.
- Lim, E. et al. (2008). Longitudinal study of the profile and predictors of left ventricular mass regression after stentless aortic valve replacement. *Annals of Thoracic Surgery*. 85(6):2026-2029.
- Parr, H., Hall, E., & Porta, N. (2022). Joint models for dynamic prediction in localised prostate cancer: a literature review. *BMC Medical Research Methodology*. 22(1):245.